

DEAD MAN'S CIPHER

MARITIME CRYPTOGRAPHIC WORKBENCH

STATUS: OPERATIONAL
Version: 1.0.0
Classification: Public Spec

1. Executive Summary & Architecture Overview

Dead Man's Cipher is a pirate-themed, enterprise-grade WebCrypto and steganography workbench built to encrypt, sign, embed, and inspect secret maritime dispatches. The application combines zero-trust browser cryptography with 17th-century naval cartography aesthetics.

System Component	Technology & Standards Used	Key Capability
Core Framework	Next.js 16 (Turbopack) + React 19 + TypeScript	Fast SSR, App Router architecture, zero-runtime overhead
WebCrypto Engine	Browser-native window.crypto.subtle	Ed25519 signatures, AES-256-GCM, PBKDF2-SHA256 (100k rounds)
Steganography Engine	Zero-Width Unicode + Canvas 2D LSB Pixels	Invisible text embedding & PNG emblem pixel data hiding
Audio Synthesizer	Web Audio API (AudioContext)	Pure JS synthesized ship bells, clicks, sirens & chimes
Audit Engine	SHA-256 Hash-Chained Immutable Log	Tamper-evident verification of all security events

2. Detailed Feature Breakdown

A. Identity Vault (Captain Keypair Management)

Provides asymmetric keypair generation using WebCrypto Ed25519/ECDSA algorithms. Each identity generates an 8-character SHA-256 public key fingerprint (e.g. 8F3A2B1C) and a printable high-resolution SVG Wax Seal emblem.

- **Key Lifecycle States:** SAILING (Active), ANCHORED (Retired), CHANGING_COLORS (Rotated), and BLACKSPOT (Revoked/Compromised).
- **Export/Import:** Portable .dmckey JSON files allow key backup and identity distribution.

B. Sealed Missive Forge (Composition & Encryption)

Encrypts secret maritime coordinates with PBKDF2-SHA256 derived keys (100,000 rounds) and symmetric AES-256-GCM / ChaCha20-Poly1305 ciphers. Every payload is digitally signed with the captain's Ed25519 private key.

- **Interactive Nautical Map:** Allows picking exact treasure coordinates directly from an interactive chart.
- **Roll Random Missive:** 1-click generator for random pirate locations, passphrases, identities, and sea shanty lyrics.

C. Boarding Inspection (4-Gate Defense Pipeline)

Intercepts dispatches and verifies authenticity across 4 mandatory security gates before decrypting secret coordinates:

Gate Check	Verification Rule	Defense Target
Gate 1: Header	Validates DMC1 magic protocol header	Protocol compliance & schema enforcement
Gate 2: Anti-Replay	Enforces 24.0-hour maximum timestamp window	Prevents replay attacks of old dispatches

Gate 3: Signature	WebCrypto Ed25519 signature verification	Blocks forged signatures & Admiralty spoofing
Gate 4: Revocation	Checks if signing key is marked BLACKSPOT	Blocks stolen or revoked compromised keys

D. Steganography Lab (Covert Carrier Channels)

Supports two distinct covert data hiding mechanisms: **Zero-Width Unicode Text Stego** (hiding bits as \u200B and \u200C in Sea Shanty lyrics) and **Canvas 2D LSB Image Stego** (storing payload bits in the Least Significant Bits of PNG emblem pixels). Includes direct NauticalMap coordinate integration.

E. Kraken's Trial Attack Simulator

Simulates 4 cyber attack scenarios (Forged Signature, Replay Attack, Blackspot Key, Ciphertext Tampering) to validate that the 4-gate defense pipeline catches and rejects malicious dispatches.

3. Security & Hash-Chained Audit Trail

All cryptographic operations produce a tamper-evident audit log entry linked via SHA-256 hashes: `currentHash = SHA256(previousHash : sequence : timestamp : eventType : actor : details)`. Users can trigger 1-click integrity verification to confirm log immutability.